Yucheng (Cecilia) Zhang

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Education

Northwestern University Sep. 2026 – May 2028 (Expected)
M.S. in Computer Science Evanston, IL

The University of Texas at Austin Aug. 2023 – May 2026
B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00 Austin, TX

Relevant Courses: Data Structures, Stochastic Processes, Elements of Software Design, Data Analytics, Database Systems, Discrete Math, Real Analysis, Probability and Statistics, Artificial Intelligence, Elements of Networking

Technical Skills

Machine Learning & AI: Large Language Models (LLMs), OpenAI API, Gemini, Claude, LangChain, CrewAI, PyTorch, TensorFlow, Scikit-learn, model evaluation (ROC/AUC)
Languages: Python, Java, JavaScript, TypeScript, SQL, R, MATLAB, SAS
Frameworks & Libraries: React, Flask, Node.js, REST APIs, Pandas, NumPy, Matplotlib, MNE, SQLAlchemy, Alembic
Databases & Cloud: PostgreSQL, SQL Server, Oracle, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI
Tools: Git, Linux, Pytest, Jest

Research Experience and Projects

CodeAssist – AI Feedback System | GitHub Aug. 2025 – Present
University of Texas at Austin | Mentor: Prof. Kia Teymourian | React, Flask, PostgreSQL, Docker Austin, TX

- Built a full-stack AI feedback system for programming courses, connecting student code editing, file uploads, submissions, and chat-based AI feedback for debugging, complexity reasoning, implementation ideas, and test-case correctness.
- Developed Flask REST APIs and PostgreSQL models for AI feedback settings, instructor controls, request limits, cooldown policies, role-based access, JSON serialization, and Alembic-backed schema changes.
- Implemented LLM context filtering so assignment descriptions, student code, uploaded files, test cases, execution results, and student output are shared only when allowed by instructor-defined policies.
- Standardized provider workflows across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, timeout handling, and assignment-level prompt behavior.
- Analyzed Docker-based autograder bottlenecks and scoped async grading with job status tracking, timing metrics, timeout handling, and isolated submission/result paths.

Researcher & TA – Biological Data Analytics Jan. 2024 – Dec. 2024
University of Texas at Austin | Mentor: Prof. Becca Young | R, Linux, AWS, Docker Austin, TX

- Processed and analyzed **250GB** of RNA-seq data using Linux, Docker, and AWS workflows, improving processing speed by 6% and reducing storage costs by 5%.
- Applied Salmon, RSEM, Trinity, and samtools for transcript quantification, achieving a **98.7% mapping rate** and ensuring reproducibility across datasets.
- Performed Voom/Limma differential-expression analysis, identified 48 key genes across time-series samples, and automated normalization, PCA, and volcano-plot workflows in R/ggplot2.
- Collaborated as a teaching assistant to train incoming FRI students on reproducible RNA-seq pipelines and data literacy.

Machine Learning Research Assistant – Neuroengineering Lab Jan. 2023 – May 2024
Remote | Mentor: Dr. Kunqiang Qing | Python, MATLAB, MNE Remote

- Processed **35GB** of multimodal EEG/fNIRS data for signal cleaning, visualization, feature extraction, connectivity analysis, and statistical testing.
- Applied statistical tests and feature-extraction methods to evaluate cognitive impairment patterns and emotion-recognition signals across experimental conditions.
- Evaluated RF, SVM, KNN, and XGBoost models, contributing to three peer-reviewed papers [2–4], including a first-author *Frontiers in Neuroscience* article and a sole-author IEEE ICIPCA paper.

Kiwifruit Quality Grading Project – Machine Learning Member Aug. 2022 – May 2024
Huazhong Agricultural University | Mentor: Prof. Yan Tong | Python, Computer Vision Wuhan, China (Hybrid)

- Contributed to an automated fruit grading pipeline using color, texture, shape, and feature-fusion methods; benchmarked model variants and analyzed results for the ML methodology section.
- Compared feature combinations to improve grading accuracy and interpretation for research later published in *Expert Systems with Applications* [1].
- Supported result interpretation for the published study by organizing feature-level evidence around image-based kiwifruit quality grading.

Work Experience

Enterprise Risk Analyst Intern May 2025 – Dec. 2025
Toyota Industries Commercial Finance, Inc. | Python, SQL Server, Azure, Power BI Dallas, TX

- Developed an end-to-end credit risk analytics pipeline integrating PayNet inquiries and internal loan servicing data through SQL Server ETL, borrower and asset feature engineering, Python modeling, and monthly reporting automation.
- Validated models with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** across Class 8 trucks, trailers, and used-vehicle loans and improving high-risk case identification by approximately **3–4%**.
- Conducted backtesting by comparing predicted risk buckets against observed outcomes, revealing up to **3.7× higher default rates** in high-risk groups and informing credit scorecard segmentation refinements.
- Automated recurring model refreshes and Power BI reporting workflows, reducing manual reporting effort by 10% while supporting portfolio risk monitoring across truck, trailer, and used-vehicle segments.

Financial Data Development Engineer – Fraud Detection Jul. 2024 – Aug. 2024
Caitong Securities Ltd | SQL, Oracle, Stored Procedures Hangzhou, China

- Designed and optimized SQL-driven ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% for transaction-monitoring and AML-related workflows.
- Developed a customer risk scoring engine by integrating multiple transactional and behavioral data streams, enhancing fraud detection precision and financial data monitoring.

Software Engineering Intern – Customer Analytics Nov. 2023 – Jun. 2024
Joblogic-X Corporation | Python, APIs, Git Dallas, TX (Hybrid)

- Engineered a scalable data pipeline to scrape and process XHS social-media data and integrate it with transactional order systems for customer analytics.
- Implemented modular ranking and clustering algorithms in Python, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh's customer-facing chatbot through REST APIs and maintained Git-based deployment workflows for reproducible updates.

Data Scientist Assistant Intern – Image Restoration Jul. 2023 – Sep. 2023
Software Engineering Center, Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, partitioned 3.4k images into train/validation/test sets, and achieved **80% accuracy** across 5 image categories.
- Integrated GFPGAN APIs for automated face restoration and color correction, reducing processing costs by 5% and supporting model-comparison reporting.

Publications

1 Miao, H., Wang, Z., **Zhang, Y.**, ... & Tong, Y. (2025). New kiwifruit quality grading methods based on image multi-feature fusion and ResTNet model. *Expert Systems with Applications*, 129507.

2 Xu, M., **Zhang, Y.**, Zhang, Y., Liu, X., & Qing, K. (2024). EEG biomarkers analysis in different cognitive impairment after stroke. *Frontiers in Neurology*, 15, 1358167.

3 **Zhang, Y.**, Zhang, Y., Jiang, Z., Xu, M., & Qing, K. (2023). The effect of EEG and fNIRS in digital therapy of Alzheimer's disease. *Frontiers in Neuroscience*, 17, 1269359.

4 **Zhang, Y.** (2023). EEG-Based on Emotion Recognition Using Machine Learning. *IEEE ICIPCA*, 226–232.

5 Li, D., Zhang, Z., & **Zhang, Y.** (2023). Public compassion under the mediation of suffering: Taking the videos of "Second Uncle" on social media as an example. *Media and Communication Research*, 4(5), 46–58.

Leadership and Community

Volunteer – Art From the Streets Jan. 2026 – Present
Community art programming for artists experiencing homelessness or housing insecurity Austin, TX

- Supported event preparation, artwork organization, and visitor engagement for community exhibitions featuring artists experiencing homelessness or housing insecurity.
- Assisted volunteers and visitors during public art events, helping maintain a respectful and welcoming environment for participating artists and community guests.

Leader – Austin Chinese Campus Christian Fellowship Aug. 2023 – Jun. 2026
University of Texas at Austin Austin, TX

- Organized weekly Bible studies, retreats, hiking trips, and cultural celebrations; coordinated transportation and scheduling to improve group communication and inclusion.
- Facilitated small-group discussions and welcomed new members through consistent event planning, follow-up communication, and peer support for students joining the group.